SPARKY'S SCIENCE EXPERIMENTS

Bottle Water Filter: Clean Muddy Water

Natural Science (water) · Ages 6+ · 30 minutes

Make a water filter to clean muddy water, from a plastic bottle, cloth, charcoal, sand and gravel.

Safety first

Sparky always says: Never drink the water from your filter - it can still have germs that make you sick!

- Only an adult can make water safe to drink, by boiling it for at least 1 minute.
- An adult should cut the bottle. Use cold charcoal only.

You need

- A 2-litre plastic cool-drink bottle
- A piece of cotton cloth (an old T-shirt)
- Cold braai charcoal, crushed small (not briquettes)
- Clean fine sand and small stones
- A cup of muddy water

Steps

1. An adult cuts the bottle in half. Turn the top half upside down in the bottom half, like a funnel.
2. Push the cloth into the neck so sand can't fall through.
3. Rinse the charcoal and sand with clean water. Add layers: charcoal (3 cm), then sand (5 cm), then stones on top (5 cm).
4. Slowly pour the muddy water onto the stones and watch it trickle down.
5. Compare the water that comes out with your muddy water.

If it doesn't work, try these in order

1. Water comes out black? That is charcoal dust - rinse the charcoal better.
2. Sand in the water? Push the cloth into the neck more firmly.
3. Still cloudy? Pour it through again, or add more sand.

Why it works

Each layer catches something. Stones catch big bits like leaves. Sand catches tiny bits of dirt between its grains. Charcoal is full of tiny holes, and some smells and colours stick to it (adsorption). Real filters use special 'activated' charcoal that works much better. No filter like this removes germs - they are far too small - so water must still be boiled.

Did you know? In 2018 Cape Town nearly ran out of water. South African water works also use sand filters, then add chlorine to kill germs.

Ask Sparky: Which layer does the most work? Why must we still boil the water?